Cerebellar Activation Differences in Major Depressive Disorder and Controls During Emotional Music and Non-Music Stimuli: An fMRI Study

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Introduction

Research shows that major depressive disorder (MDD) can alter the way individuals assess the emotional valence and arousal of music (Gui et al., 2019; Lepping et al., 2016). Studies found that MDD patients show atypical patterns, like a blunted or reversed brain response, particularly in regions associated with emotion regulation such as the anterior cingulate and dorsolateral prefrontal cortex (Lepping et al., 2016). The cerebellum, long viewed as a motor structure, has been increasingly recognized for its crucial involvement in emotion processing (Pierce et al., 2023). Nevertheless, the role of the cerebellum in music emotion recognition has not been sufficiently elucidated. Therefore, this study focused on the cerebellum to examine cerebellar activity during arousal and valence evaluation while listening to music. This study aimed to compare cerebellar activation patterns between Control and MDD subjects during emotional music and non-music stimuli using fMRI data from the ds000171 dataset. By performing trial-wise activation analysis and group comparisons. The results showed significant difference for positive non-music trials, where 'Control subjects' had higher cerebellar activation than 'MDD subjects'.

Methods

Dataset

- The dataset (ds000171) came from OpenNeuro, included 39 participants (19 MDD, 20 Controls).
- During fMRI scanning, participants listened to emotional auditory stimuli in two categories: musical and non-musical. Each participant completed five runs; three music runs and two non-music runs

First-level analysis: Extraction of trial-wise cerebellar activation

- Extracted BOLD signals from cerebellum ROI using a probabilistic cerebellum atlas
- Cleaned signals using motion-related confounds (6 parameters: translation & rotation)
- Computed mean BOLD signal across cerebellar voxels at each time point
- Trial-level activation: averaged cerebellar signal within the onset duration window for each trial (positive or negative music/nonmusic)

Second-level analysis: Group comparison using two-sample t-test

- Trial-level activations aggregated by group (Control vs. MDD)
- Two-sample t-tests conducted separately for each emotion × sound type condition ($p < 0.05$)
- Averaged cleaned cerebellar voxel signals across relevant trials for each subject/run and econstructed voxel-wise activation maps in brain space.

Discussion

The results showed no significant differences in cerebellar activation between Control and MDD groups during positive or negative music trials, suggests there is no evidence that emotional music differentially engaged the cerebellum. However. for positive non-music stimuli, Controls exhibited higher cerebellar activation than individuals diagnosed as MDD. These findings align with previous research showing altered cerebellar activation in MDD during emotional and cognitive processing tasks (Depping et al., 2018; Wu et al., 2024), indicating that non-musical positive stimuli may engage the cerebellum more strongly in healthy individuals. Future studies could further study would beneficial to investigate whether music and non music emotional auditory stimuli processed differently, and how cerebellar responses differ across a range of emotional non-music stimuli to better understand emotion processing in MDD.

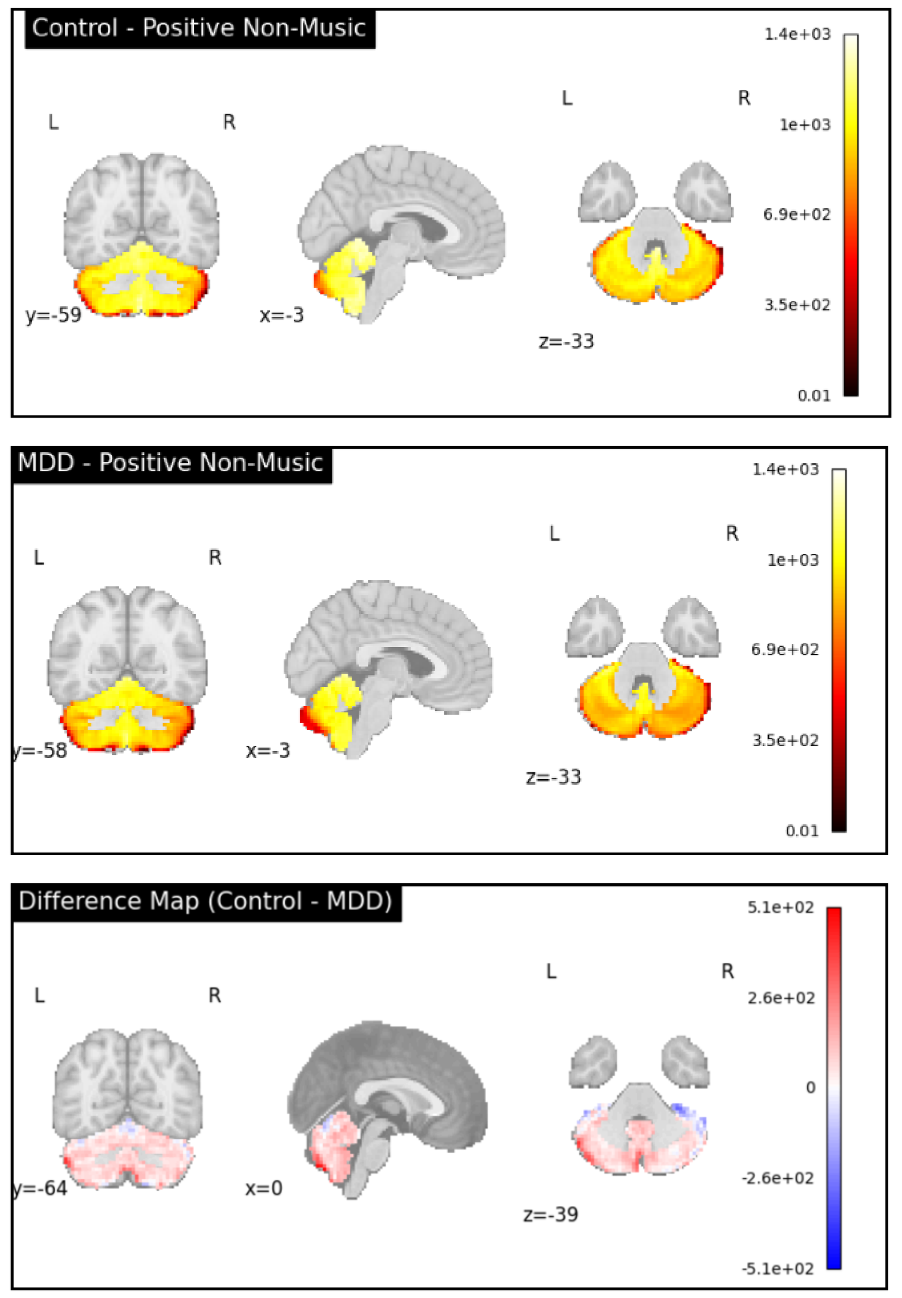
Results

Cerebellar activation was compared between Control and MDD groups across four conditions: positive and negative music, and positive and negative non-music.

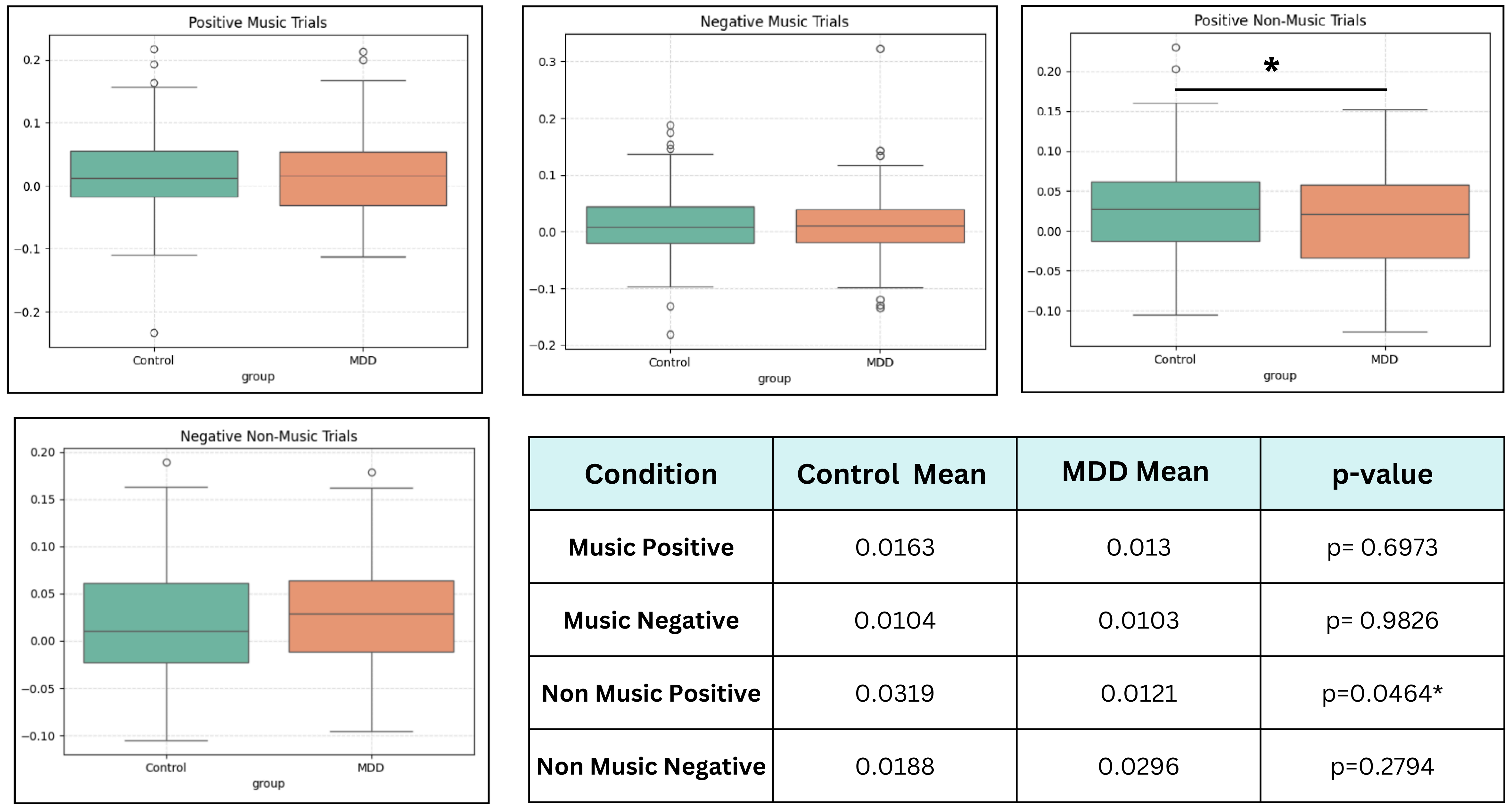
- Music trials: No significant differences were observed between the groups for either positive or negative music.
- Non-music trials: A significant difference was found in positive non-music trials, with higher activation in the Control group. No significant difference was observed in negative non-music trials.

Cerebellar Activation Maps

Control vs. MDD



Per-Condition analysis



Condition	Control Mean	MDD Mean	p-value
Music Positive	0.0163	0.013	p= 0.6973
Music Negative	0.0104	0.0103	p= 0.9826
Non Music Positive	0.0319	0.0121	p=0.0464*
Non Music Negative	0.0188	0.0296	p=0.2794

- Adamaszek, M., Manto, M., & Schutter, D. J. L. G. (2022). Introduction into the Role of the Cerebellum in Emotion. Advances in Experimental Medicine and Biology, 3–12. https://doi.org/10.1007/978-3-030-99550-8_1
- Boayue, N. M., Csifcsák, G., Puonti, O., Thielscher, A., & Mittner, M. (2018). Head models of healthy and depressed adults for simulating the electric fields of non-invasive electric brain stimulation. F1000Research, 7, 704. https://doi.org/10.12688/f1000research.15125.2
- Depping, M. S., Schmitgen, M. M., Kubera, K. M., & Wolf, R. C. (2018). Cerebellar contributions to Major Depression. Frontiers in Psychiatry, 9. https://doi.org/10.3389/fpsyt.2018.00634
- Lepping, R. J., Atchley, R. A., Chrysikou, E., Martin, L. E., Clair, A. A., Ingram, R. E., Simmons, W. K., & Savage, C. R. (2016). Neural processing of emotional musical and nonmusical stimuli in depression. PLoS ONE, 11(6), e0156859. https://doi.org/10.1371/journal.pone.0156859
- Pierce, J. E., Thomasson, M., Voruz, P., Selosse, G., & Peron, J. (2023). Explicit and implicit emotion processing in the cerebellum: a meta-analysis and systematic review. The Cerebellum, 22(5), 852-864.
- Lepping, R. J., Atchley, R. A., Chrysikou, E., Martin, L. E., Clair, A. A., Ingram, R. E., ... & Savage, C. R. (2016). Neural processing of emotional musical and nonmusical stimuli in depression. PloS one, 11(6), e0156859.
- Gui, R., Chen, T., & Nie, H. (2019). The Impact of Emotional Music on Active ROI in Patients with Depression Based on Deep Learning: A Task-State fMRI Study. Computational Intelligence and Neuroscience, 2019(1), 5850830.

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